Ch 5.1.5: k-fold Cross-Validation for Classification

Lecture 15 - CMSE 381

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Mon, Oct 9, 2023

Announcements

Last time:

- k-fold CV

This lecture:

- CV for classification

Announcements:

- Homework #4 is posted, Due tonight
- Grades

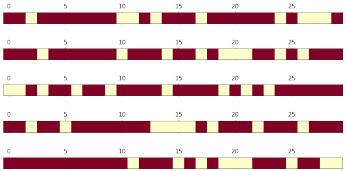
Percent	Convert
≥ 90%	4.0
≥ 85%	3.5
≥ 80%	3
≥ 75%	2.5
≥ 70%	2
≥ 65%	1.5
≥ 60%	1
< 60%	0

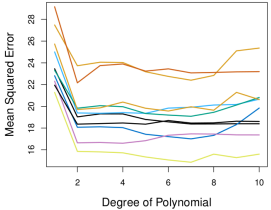
Section 1

Last time

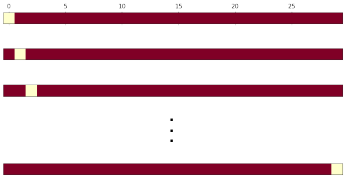
Approximations of Test Error

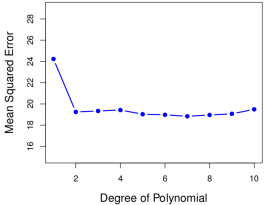
Validation Set



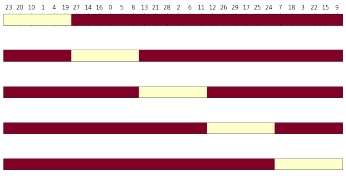


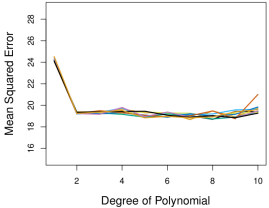
LOOCV





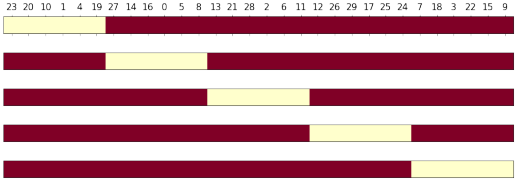
K-fold CV





Definition of k-fold CV — Regression

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points



- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell}\sum_{(x_j,y_j)\in U_i}(y_j - \hat{y}_j)^2$
- Rinse and repeat

Return

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{MSE}_i$$

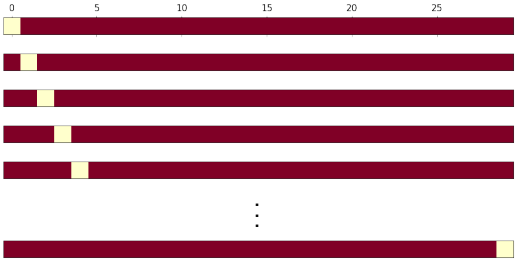
Section 2

CV for Classification

Setup: LOOCV for Classification

- Remove ith point (x_i, y_i) and reserve for testing.
- Train the model on remaining points
- Calculate $\boxed{\mathrm{Err}_i = \mathrm{I}(y_j \neq \hat{y}_j)}$
- Rinse and repeat

$$\mathrm{I}(y_j \neq \hat{y}_j) = \begin{cases} 1 & \text{if } y_j \neq \hat{y}_j \\ 0 & \text{else } (y_j = \hat{y}_j) \end{cases}$$

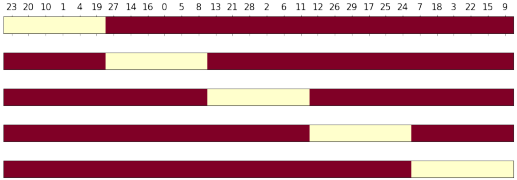


Return

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{Err}_i$$

Setup: k-fold

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points



- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{Err}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} \underbrace{\mathrm{I}(y_j \neq \hat{y}_j)}$

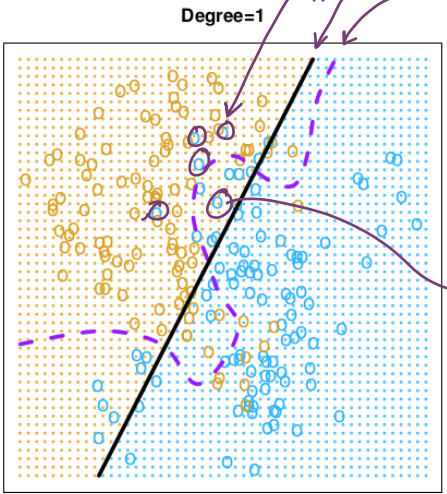
  # of elts in U_i 1 if $y_i \neq \hat{y}_i$
- Rinse and repeat

Return

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{Err}_i$$

Example on simulated data: Linear

Incorrect for Bayes and for Logistic regression



where my prediction switches

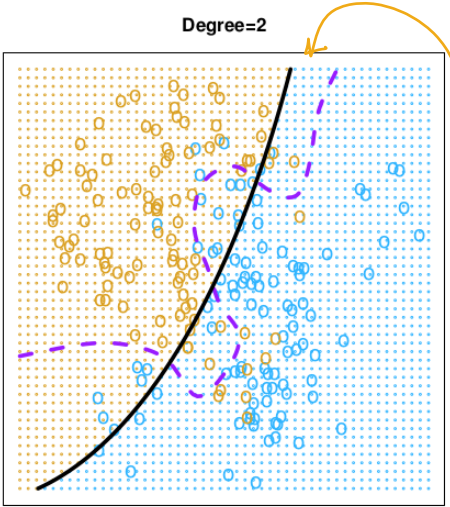
- Purple: Bayes decision boundary.
  - Error rate: 0.133 13%
- Black: Logistic regression
  - $\log(p/(1-p)) = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2}_{20\%}$
  - Error rate: 0.201

Correct for Bayes

Incorrect for Log reg.

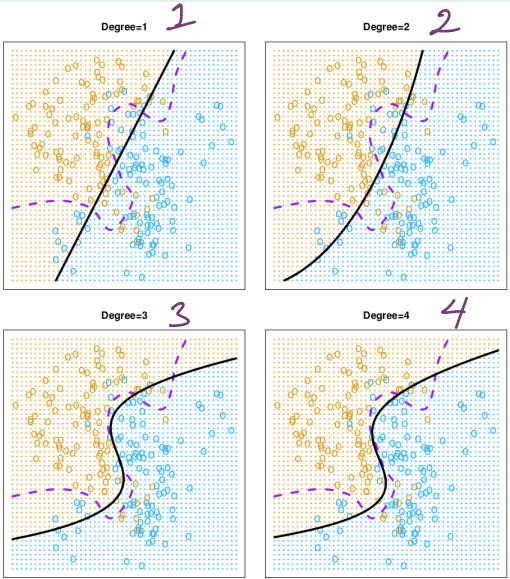
More flexibility by replacing with polynomial

Example on simulated data: Quadratic logistic regression



- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - $\log(p/(1-p)) = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + \beta_4 X_2^2$
  - Error rate: 0.197

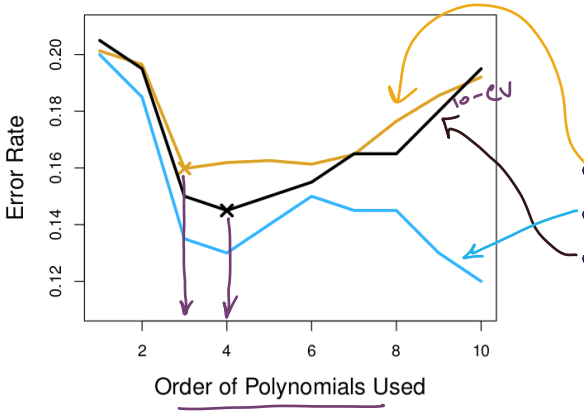
Example on simulated data: all the polynomials!



- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - Deg 1 Error rate: 0.201
  - Deg 2 Error rate: 0.197
  - Deg 3 Error rate: 0.160
  - Deg 4 Error rate: 0.162

Better accuracy

Decide degree based on CV



more flexibility

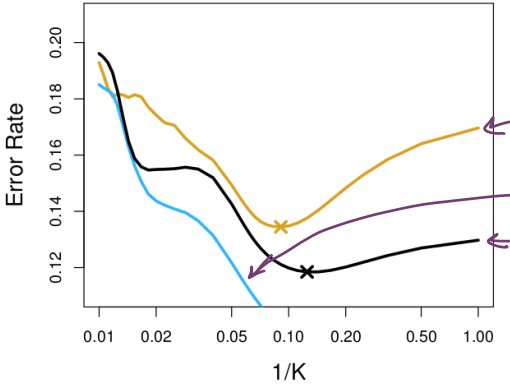
- Test error (brown) U-shape
- Training error (blue) monotone decreasing
- 10-fold CV error (black)

Under approx of test error

Close location of minimum

$\Rightarrow$ Choose the model w/ degree 4

Similar game for KNN

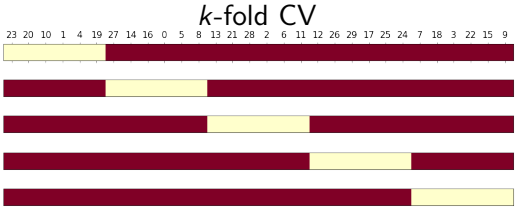


- Test error (brown)
- Training error (blue)
- 10-fold CV error (black)

Under estimate
But the chosen k
is close

Coding - k-fold for penguin classification section

TL;DR



$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{MSE}_i$$

Use $k = 5$ or 10 usually

k-fold CV for classification

$$\mathrm{Err}_i = \mathrm{I}(y_j \neq \hat{y}_j)$$

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{Err}_i$$

Next time

12	Mon	Oct 2	Leave one out CV	5.1.1, 5.1.2	
13	Wed	Oct 4	k-fold CV	5.1.3	
14	Fri	Oct 6	More k-fold CV,	5.1.4-5	
15	Mon	Oct 9	k-fold CV for classification	5.1.5	HW #4 Due
16	Wed	Oct 11	Resampling methods: Bootstrap	5.2	
17	Fri	Oct 13	Subset selection	6.1	
18	Mon	Oct 16	Shrinkage: Ridge	6.2.1	
19	Wed	Oct 18	Shrinkage: Lasso	6.2.2	
	Fri	Oct 20	**Review**		
	Mon	Oct 23	No class - Fall break		
	Wed	Oct 25	**Midterm #2**		
20	Fri	Oct 27	Dimension Reduction	6.3	